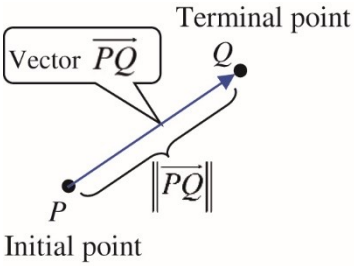
Interpret Vectors Geometrically

Scalars: Quantities that are described by a single value called the magnitude (the size or measure of the quantity)
Vectors: Quantities that are described by both magnitude *and* direction

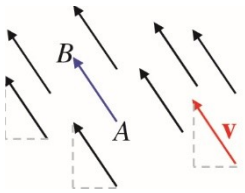
Example: The *speed* of an object is a scalar quantity that tells us how fast an object travels.
The *velocity* of an object is a vector quantity, because it tells us how fast the object travels *and* in which direction.

A vector in a plane is a line segment with a specified direction.



Vector $\overrightarrow{PQ}$ is an arrow with endpoints P and Q pointing from P to Q.
P is the initial point of the vector and Q the terminal point of the vector.
The magnitude of $\overrightarrow{PQ}$, denoted $\|\overrightarrow{PQ}\|$, is the length of the line segment $\overrightarrow{PQ}$.

Using the context of the displacement of an object, the vector refers to the object's overall change in position in moving from point P to point Q.
Two vectors are equal if they have the same magnitude and direction.



The vectors in the figure are all equal to $\overrightarrow{AB}$.
The vectors each represent the same displacement left and upward from the initial point to the terminal point.

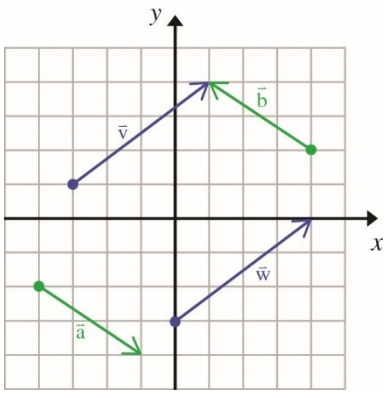
For vectors, we often use boldface lower case letters such as **u**, **v**, and **w**.

If **v** = $\overrightarrow{AB}$, then **v** represents the set of *all* vectors equal to $\overrightarrow{AB}$.

Because it is difficult to depict handwritten boldface, we sometimes use an arrow over a lowercase letter: $\vec{v}$

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1. Determine if the two vectors are equal.



v: from $(-3,1)$ to $(1, 4)$ **w**: from $(0,-3)$ to $(4, 0)$

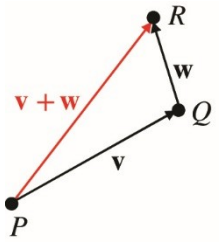
Yes

a: from $(-4,-2)$ to $(-1,-4)$ **b**: from $(4, 2)$ to $(1, 4)$

No; wrong direction

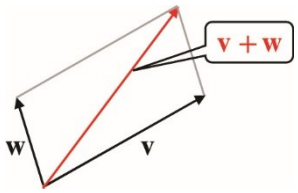
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If vector **v** represents the displacement from P to Q and vector **w** represents the displacement from Q to R, then the vector **v + w** represents the displacement from P to R.



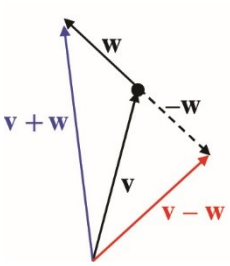
The vector **v** + **w** is formed by placing **v** and **w** "tip to tail," (the initial point of **w** is placed at the terminal point of **v**).

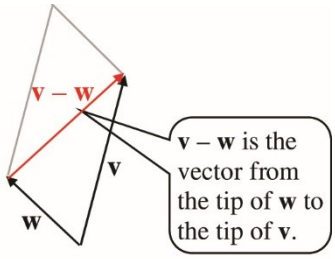
The sum **v** + **w** can also be visualized by placing **v** and **w** at a common initial point, "tail" to "tail".
The sum **v** + **w** is the diagonal of the parallelogram from the initial point of **v** and **w** to the opposite vertex.



The vector −**v** is −1**v** and represents the opposite of **v** (same magnitude, but opposite direction).

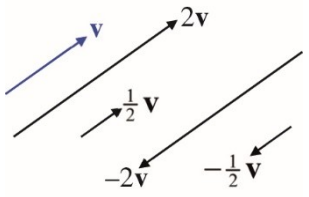
We use the opposite of a vector to define vector subtraction.
The difference of vectors **v** and **w** is given by $\mathbf{v - w = v + (-w)}$.





If a vector **v** is multiplied by a real number k, the magnitude of $k\mathbf{v}$ is $|k|\|\mathbf{v}\|$ (the absolute value of k times the magnitude of **v**).

The direction of $k\mathbf{v}$ is in the same direction as **v** if $k > 0$ and in the opposite direction as **v** if $k < 0$.



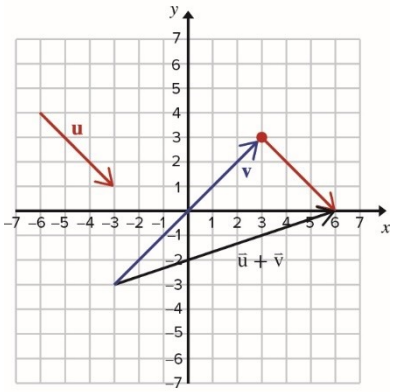
The value k is a scalar, and for this reason, the vector $k\mathbf{v}$ is called a scalar multiple of **v**.

The value of k either stretches or shrinks the length (magnitude) of **v** by a factor of $|k|$.
If $k = 0$, then $k\mathbf{v}$ is the zero vector, denoted **0**.
The zero vector has a magnitude of 0 and is assigned no direction.

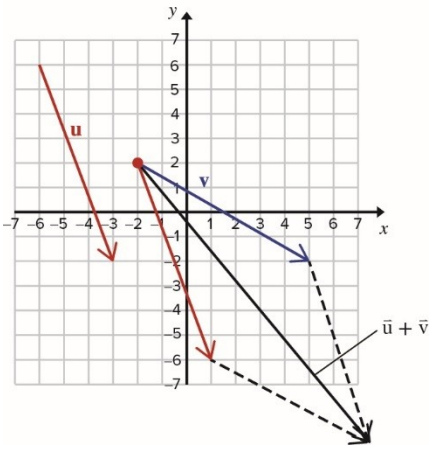
A vector sketched with initial point at the origin of a rectangular coordinate system is said to be drawn in standard position.

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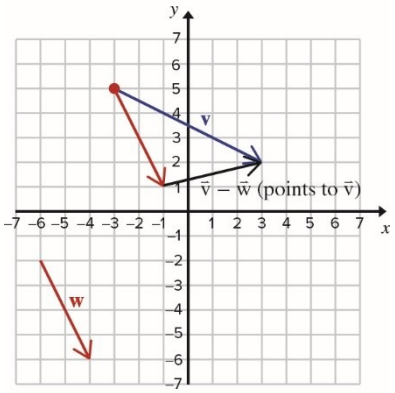
2. Sketch **u** + **v** by placing the initial point of **u** at the terminal point of **v**.



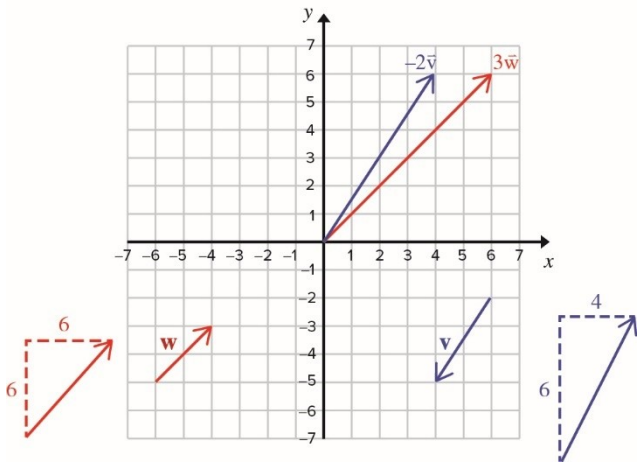
3. Sketch **u** + **v** by drawing **u** and **v** with the same initial point.



4. Sketch **v** − **w** by drawing **v** and **w** with the same initial point.



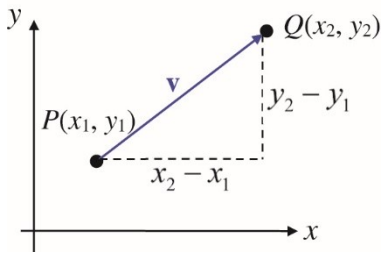
5. Sketch −2**v** and 3**w** in standard position.



Represent Vectors in Component Form

$P(x_1, y_1)$ $Q(x_2, y_2)$

Let and be points in the plane, and let **v** = $\overrightarrow{PQ}$. The component form of **v** is given by



$$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$$

Note that vector notation $\langle a, b \rangle$ is *not* the same as point notation (a, b).

---

① ②

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6. Given vector **v** with initial point $(3,8)$ and terminal point $(-1,3)$, find the component form of **v.**

$v = \langle -1-3, 3-8 \rangle = \langle -4, -5 \rangle$

7. Given vector **v** with initial point $(12,-10)$ and terminal point $(-20, 10)$, find the component form of **v**.

$v = \langle -20-12, 10-(-10) \rangle = \langle -32, 20 \rangle$

If **v** is placed with initial point at $(5, 3)$, what is the terminal point of **v**?

terminal point $(5-32, 3+20)$

$= (-27, 23)$

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Perform Operations on Vectors in Component Form

$\langle a_1, b_1 \rangle$ $\langle a_2, b_2 \rangle$, $a_1 = a_2$ $b_1 = b_2$.

Equality of Vectors If **v** = and **w** = then **v** = **w** if and only if and

$= \langle a_1, b_1 \rangle$, $\langle a_2, b_2 \rangle$,

Operations on Vectors Let **v** **w** = and c be a real number.

Addition $\mathbf{v} + \mathbf{w} = \langle a_1 + a_2, b_1 + b_2 \rangle$ **Subtraction** $\mathbf{v} - \mathbf{w} = \langle a_1 - a_2, b_1 - b_2 \rangle$

Multiplication of a vector by a scalar $c\mathbf{v} = \langle ca_1, cb_1 \rangle$

Properties of Vectors If **u**, **v**, and **w** are vectors, and c and d are scalars, then

1. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$
2. $(\mathbf{v} + \mathbf{w}) + \mathbf{u} = \mathbf{v} + (\mathbf{w} + \mathbf{u})$
3. $\mathbf{v} + \mathbf{0} = \mathbf{v}$
4. $\mathbf{v} + (-\mathbf{v}) = 0$
5. $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$
6. $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$
7. $(cd)\mathbf{v} = c(d\mathbf{v})$
8. $1(\mathbf{v}) = \mathbf{v}$
9. $0(\mathbf{v}) = \mathbf{0}$
10. $c(\mathbf{0}) = \mathbf{0}$
11. $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$

$\langle 1, -4 \rangle$ $\langle 2, 3 \rangle$,

Given **r** = and **s** = find

8. $\mathbf{r} + \mathbf{s}$

$= \langle 1 + 2, -4 + 3 \rangle$

$= \langle 3, -1 \rangle$

9. $-4\mathbf{s}$

$= -4\langle 2, 3 \rangle$

$= \langle -8, -12 \rangle$

10. $3\mathbf{r} + 2\mathbf{s}$

$= 3\langle 1, -4 \rangle + 2\langle 2, 3 \rangle$

$= \langle 3, -12 \rangle + \langle 4, 6 \rangle$

$= \langle 7, -6 \rangle$

Represent Vectors in Terms of **i** and **j**

$\mathbf{v} = \langle a, b \rangle$: $\|\mathbf{v}\| = \sqrt{a^2 + b^2}$

Unit vector: Vector with a magnitude of 1 **Magnitude of**

To find a Unit Vector in the Direction of a Given Vector:

If $\mathbf{v} = \langle a, b \rangle$, then a unit vector $\mathbf{u}_v$ in the direction of **v** is given by $\mathbf{u}_v = \frac{1}{\|\mathbf{v}\|}\mathbf{v} = \frac{1}{\|\mathbf{v}\|}\langle a, b \rangle = \left\langle \frac{a}{\|\mathbf{v}\|}, \frac{b}{\|\mathbf{v}\|} \right\rangle$.

---

11. Find a unit vector $\mathbf{u}_w$ in the direction of $\mathbf{w} = \langle -8, 15 \rangle$ and verify that it is a unit vector.

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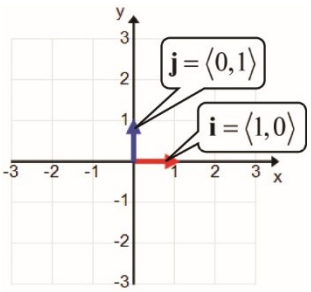
$$\left\|\vec{w}\right\| = \sqrt{(-8)^2 + (15)^2} = 1$$

$$\vec{u}_{\vec{w}} = \frac{1}{17}\langle -8, 15\rangle = \left\langle -\frac{8}{17}, \frac{15}{17}\right\rangle$$

$$\left\|\vec{u}_{\vec{w}}\right\| = \sqrt{\left(\frac{-8}{17}\right)^2 + \left(\frac{15}{17}\right)^2} = 1 \quad \therefore \vec{u}_{\vec{w}} \text{ is a unit vector.}$$

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Two important unit vectors are $\mathbf{i} = \langle 1,0 \rangle$ and $\mathbf{j} = \langle 0,1 \rangle$ because they are unit vectors in the direction of the positive *x*- and *y*-axes, respectively.



The representation of $\mathbf{v} = \langle a,b \rangle$ in terms of **i** and **j** is $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$.

The values a and b are called the scalar horizontal and scalar vertical components of **v**.

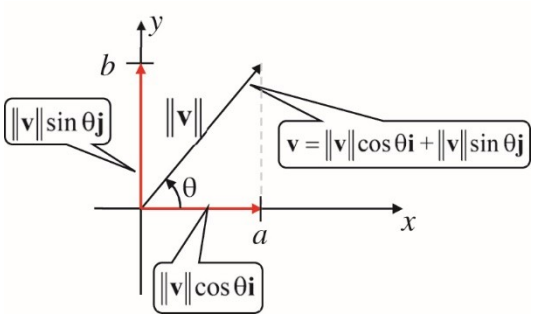
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12. Write $\langle -12, -30 \rangle$ in terms of **i** and **j**. $-12\mathrm{i} - 30\mathrm{j}$ note: i is not i

---

Relate Component Form and Direction and Magnitude of a Vector

Given a vector in standard position, its direction angle θ is measured counterclockwise from the positive *x*-axis over the interval $0 \le \theta < 2\pi$ or $0° \le \theta < 360°$.



$0° \le \theta < 360°$ be the direction of **v** measured counterclockwise from the positive *x*-axis.

$\|\mathbf{v}\| = \sqrt{a^2 + b^2}$ and $\tan\theta = \dfrac{b}{a}$ $(a \neq 0)$

(magnitude and direction of **v**)

$a = \|\mathbf{v}\|\cos\theta$ and $b = \|\mathbf{v}\|\sin\theta$

(scalar components of **v**)

$\mathbf{v} = \langle a,b \rangle = \langle \|\mathbf{v}\|\cos\theta, \|\mathbf{v}\|\sin\theta \rangle$ or $\mathbf{v} = a\mathbf{i} + b\mathbf{j} = \|\mathbf{v}\|\cos\theta\mathbf{i} + \|\mathbf{v}\|\sin\theta\mathbf{j}$

Magnitude, Direction, and Components of a Vector:

Let $\mathbf{v} = \langle a,b \rangle$ be a vector in standard position, and let

---

13. Write the vector **v** in the form $a\mathbf{i} + b\mathbf{j}$ where **v** has magnitude $\|\mathbf{v}\| = 10$ and direction angle $\theta = \dfrac{3\pi}{4}$.

$$a = 10\cos\frac{3\pi}{4} = -5\sqrt{2}$$
$$b = 10\sin\frac{3\pi}{4} = 5\sqrt{2}$$

$\vec{\mathrm{v}} = -5\sqrt{2}\,\mathrm{i} + 5\sqrt{2}\,\mathrm{j}$

14. Find the magnitude and direction angle ($0 \le \theta < 360°$) for $\mathbf{s} = 7\mathbf{i} - 3\mathbf{j}$. Round to 1 decimal place.

$\|\vec{\mathrm{s}}\| = \sqrt{7^2 + (-3)^2} = \sqrt{58}$

$\|\mathrm{s}\| \approx 7.6$

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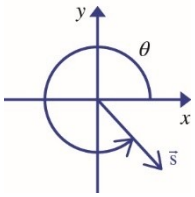
$\tan\theta = -\frac{3}{7}$,

$\theta = \tan^{-1}\left(-\frac{3}{7}\right) + 360°$

$\theta \approx -23.2° + 360°$

$\theta \approx 336.8°$

s in QIV



15. A baseball is thrown with a speed of 65 mph at an angle of 12° from the horizontal. Write the velocity vector **v** at the time of release in terms of **i** and **j**. Round the components to the nearest tenth of a mph.



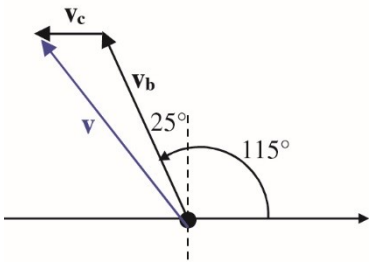
$\|v\| = 65$ $\quad a = 65\cos 12° \approx 63.6$

$\theta = 12°$ $\quad b = 65\sin 12° \approx 13.5$

$v = ai + bj$ $\quad v = 63.6i + 13.5j$

Use Vectors in Applications

16. A boat travels 10 knots at a bearing of N25°W in still water. It encounters a current moving due west at 3 knots. Find the true speed (to 1 decimal place) and direction of the boat (to the nearest degree).



The velocity of the boat in still water is given by:

$$v_b = (10\cos 115°)i + (10\sin 115°)j$$

The velocity of the current is given by:

$$v_c = -3i$$

The net velocity of the boat is the sum of:

$$\begin{aligned} v &= v_b + v_c \\ &= (10\cos 115°)i + (10\sin 115°)j + (-3i) \\ &= (10\cos 115° - 3)i + (10\sin 115°)j \\ &\approx -7.2i + 9.1j \end{aligned}$$

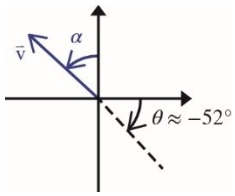
$$\text{true speed} = \|v\| = \sqrt{(-7.2)^2 + 9.1^2}$$

$$\approx \boxed{11.6 \text{ knots}}$$

$$\text{direction of boat } \tan\theta \approx \frac{9.1}{-7.2}$$

$$\theta \approx \tan^{-1}\left(-\frac{9.1}{7.2}\right) \approx -52°$$

$$\alpha = 90° - 52° = 38°$$



The true speed is 11.6 knots in the direction N38°W.

---

17. The forces (in units of pounds) $\mathbf{F_1} = 10\mathbf{i} - 3\mathbf{j}$, $\mathbf{F_2} = -2\mathbf{i} + \mathbf{j}$, and $\mathbf{F_3} = 11\mathbf{i}$ act on an object.

a. Find the resultant force, **R**.

$$\begin{aligned} R &= F_1 + F_2 + F_3 \\ &= (10i - 3j) + (-2i + j) + 11i \\ &= 19i - 2j \end{aligned}$$

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b. What additional force **F** is needed for the object to be in static equilibrium?

$$F = -19i + 2j$$

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